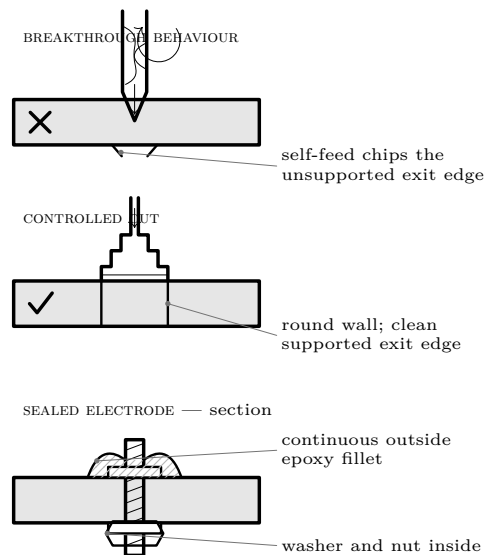


Craft 5: Drill & Seal

Every hole you make is a hole in a pressure vessel

THE STANDARD

A round hole of the intended size, with no cracks radiating from it, no crazing around it, and a seal that is continuous all the way round on the **outside** face. Nothing you fit through a chamber wall may rely on the threads alone to hold pressure.



PVC grabs

A standard twist drill's cutting geometry pulls itself into plastic. As the point breaks through, the flutes snatch, and the drill either climbs through violently or cracks the far wall. This surprises everyone once.

Use a step drill

It cannot self-feed, it leaves a clean round hole, and it works from a small pilot upward. This is the right tool and it is inexpensive.

Or grind the drill

If you only have twist drills, back the cutting edge off to near zero rake, or drill a pilot and then open up in stages. Slow speed, light steady pressure, and back off as you approach breakthrough.

Back it up

Clamp a scrap block behind the wall where you can. It supports the material at the moment of breakthrough, which is the moment things go wrong.

Never near a joint

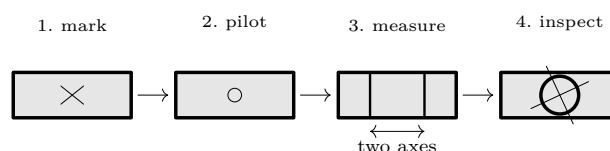
Keep penetrations well away from a welded socket. The wall there is a different thickness and the weld zone has different properties.

Round holes only

No slots, no filed-out ovals, no drilling two overlapping holes. Every corner is a stress concentration and a crack starter.

Method for the igniter electrodes

1. Mark both positions from the drawing, well clear of any welded joint. Write the specified hole diameter beside each cross. Measure the distance to the nearest socket edge and record it before cutting. *see the **Craft 1: Measure & Mark** sheet*
2. Pilot-drill small and slow. Stop and measure the pilot centre from the same datum. If it is not where the drawing puts it, remake the part; do not steer, slot or overlap the hole.
3. Open to size with a step drill, backing off pressure at breakthrough.
4. Measure the finished hole in two directions, 90° apart. Both readings must match the drawing and each other. An oval or oversize hole is a remake. Deburr both sides. *see the **Craft 3: Deburr & Chamfer** sheet*
5. Fit the screw with a washer inside and out, and a nut inside. Do not overtighten — PVC creeps, and a nut torqued hard today is loose next month.
6. Set the internal electrode gap at 3 to 5 mm and **test the spark before sealing anything**, with the chamber open.
7. Seal the outside face with an epoxy fillet all the way round the screw. The epoxy is the pressure seal; the nut only holds position.
8. Let the epoxy cure fully before the chamber sees any pressure.



FOUR STOPS BEFORE SEALANT

After marking: Is the centre referenced to a datum, not guessed? **After the pilot:** Is it still on the cross? **After opening:** Do two diameter readings agree? **After deburring:** Does raking light show a clean, round edge with no white lines? Record each answer. A doubtful answer means remake, not repair.

TEST BEFORE YOU SEAL

Once the epoxy is on, changing the gap means destroying the seal. Test the spark with a battery-free piezo click while you can still reach both electrodes, and confirm you see and hear it jump. Sealing an untested igniter is how a build ends up needing a new chamber.

Why a crack near a hole is different

A crack in plain pipe wall has to start somewhere. A hole gives it a start: the stress in the material flows around the hole and concentrates at its edge, roughly tripling there even for a perfectly round hole. A sharp corner, a scratch from a slipped drill, or a star crack from a snatching bit multiplies that far further. This is why the rules are round holes, no sharp corners, no scratches, and inspect every penetration before every session.

Check it yourself. Hold the drilled area up to a strong light and look at the plastic around the hole from several angles. You are looking for **crazing** — a faint web of tiny white lines — and for any crack, however short. Crazing means the material was stressed while you drilled it. Either is a reason to remake the part, not to seal over it.

Final proof

Before accepting the part, make one record containing the drawing diameter, both measured diameters, distance from the chosen datum, cure start time and cure-complete time. Add one close photograph in raking light before sealant and one after cure, with the complete fillet visible. Initial the record. This is proof that the checks happened; it is not permission to test with pressure.

The habit worth taking away

Match the tool to the material rather than forcing the tool you have. A twist drill is a superb tool for steel and a poor one for thin plastic, and no amount of care fully compensates. Recognising that a job wants a different tool — and that the right one often costs very little — is most of what separates work that looks made from work that looks got away with.

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